

Construction Robotics Competency Regimes

Five dominant failure-physics evidence domains

1 Earthwork and Material Processing

Evolving soil resistance, terrain interaction, and material heterogeneity



48.0%

n=36



2 Structural Assembly and Installation

Contact jamming, tolerance amplification, and insertion sensitivity



24.0%

n=18



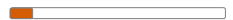
3 Material Placement and Lifting

Suspended-load dynamics and placement uncertainty



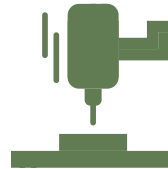
10.7%

n=8



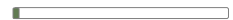
4 Additive Manufacturing and Surface Processing

Cumulative process-state drift and deposition or coverage consistency



2.7%

n=2



5 Navigation, Layout, and Logistics Support

Mixed-agent flow, occlusion, path uncertainty, and human co-presence



14.7%

n=11

